



AIST Tutorial: How to use and program Gemini

Introduction to QuEra



QuEra's Mission:

Accelerate innovation by building useful, scalable,
fault-tolerant quantum computers

Introduction to QuEra



Core
technology

Analog:
Hamiltonian
programming

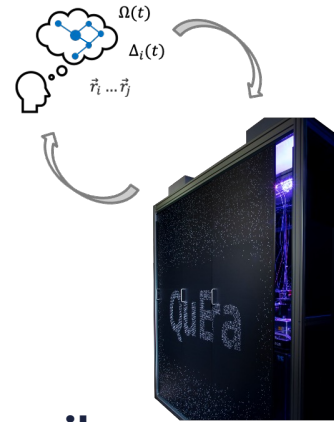
Digital (NISQ):
quantum circuit
& error correction
programming
w/ physical qubits

Digital (fault-tolerant):
quantum circuit
programming
w/ logical qubits

Introduction to QuEra



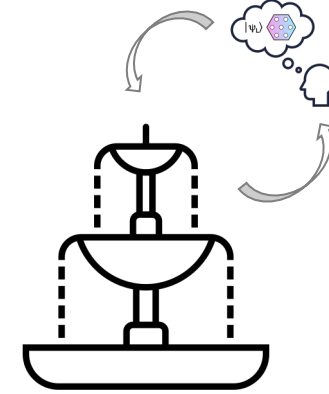
Introduction to QuEra



- **Aquila**
- Analog Hamiltonian Simulation (AHS)
- Cloud service



- **Gemini**
- On-prem installations
- HW-SW-APP co-design



- **'M4'** class machines
- Logical QPUs as OEM devices
- Business solutions

Products & services

Software

APIs, SDK & cloud

Compilers & HPC

Decoding & applications

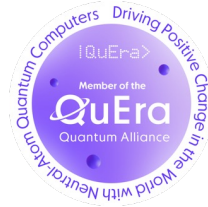
Core technology

Analog

Digital (NISQ)

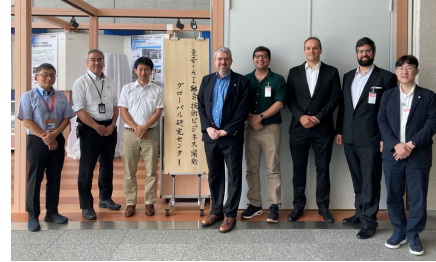
Digital (fault-tolerant)

Introduction to QuEra



Key partners

- AWS
- Harvard & MIT
- LANL & NERSC
- DOD and DOE
- QuEra Quantum Alliance



- AIST (Japan)
- NQCC (UK)
- Google QuantumAI

- DARPA QBI
- NVIDIA
- Private-public ecosystems in Japan and UK
- Further global expansion

Products & services

Aquila/AHS @cloud

Gemini on-prem & co-design

Logical QPUs & solutions

Software

APIs, SDK & cloud

Compilers & HPC

Decoding & applications

Core technology

Analog

Digital (NISQ)

Digital (fault-tolerant)

The rapid progress of neutral atom computing

1st coherence-preserving transport of atoms in tweezers

Bluvstein, D., Levine, H., Semeghini, G. *et al.*
A quantum processor based on coherent transport of entangled atom arrays.
Nature **604**, 451–456 (2022).

Quantum circuits on logical qubits

Bluvstein, D., Evered, S.J., Geim, A.A. *et al.*
Logical quantum processor based on reconfigurable atom arrays.
Nature **626**, 58–65 (2024).

Continuous operation and scale

Chiu, NC., Trapp, E.C., Guo, J. *et al.*
Continuous operation of a coherent 3,000-qubit system.
Nature **646**, 1075–1080 (2025).

2-qubit Rydberg gate fidelity reaches 99.5%

Evered, S.J., Bluvstein, D., Kalinowski, M. *et al.*
High-fidelity parallel entangling gates on a neutral-atom quantum computer.
Nature **622**, 268–272 (2023).

Demonstration of all ingredients for a universal fault-tolerant neutral-atom QPU

Bluvstein, D., Evered, S.J., Geim, A.A. *et al.*
Architectural mechanisms of a universal fault-tolerant quantum computer.
arXiv:2506.20661 (2025)

2017

1st deterministic preparation of atom arrays in tweezers; programmable Rydberg Hamiltonian in 1D

Bernien, H., Schwartz, S., Keesling, A. *et al.*
Probing many-body dynamics on a 51-atom quantum simulator.
Nature **551**, 579–584 (2017).

Programmable Rydberg Hamiltonian in 2D

Ebadi, S., Wang, T.T., Levine, H. *et al.*
Quantum phases of matter on a 256-atom programmable quantum simulator.
Nature **595**, 227–232 (2021).

QuEra's 1st-generation analog mode computer

Wurtz, J., Bylinskii A., *et. al.* Aquila: QuEra's 256-qubit neutral-atom quantum computer.
Arxiv:2306.11727

2025

digital architecture

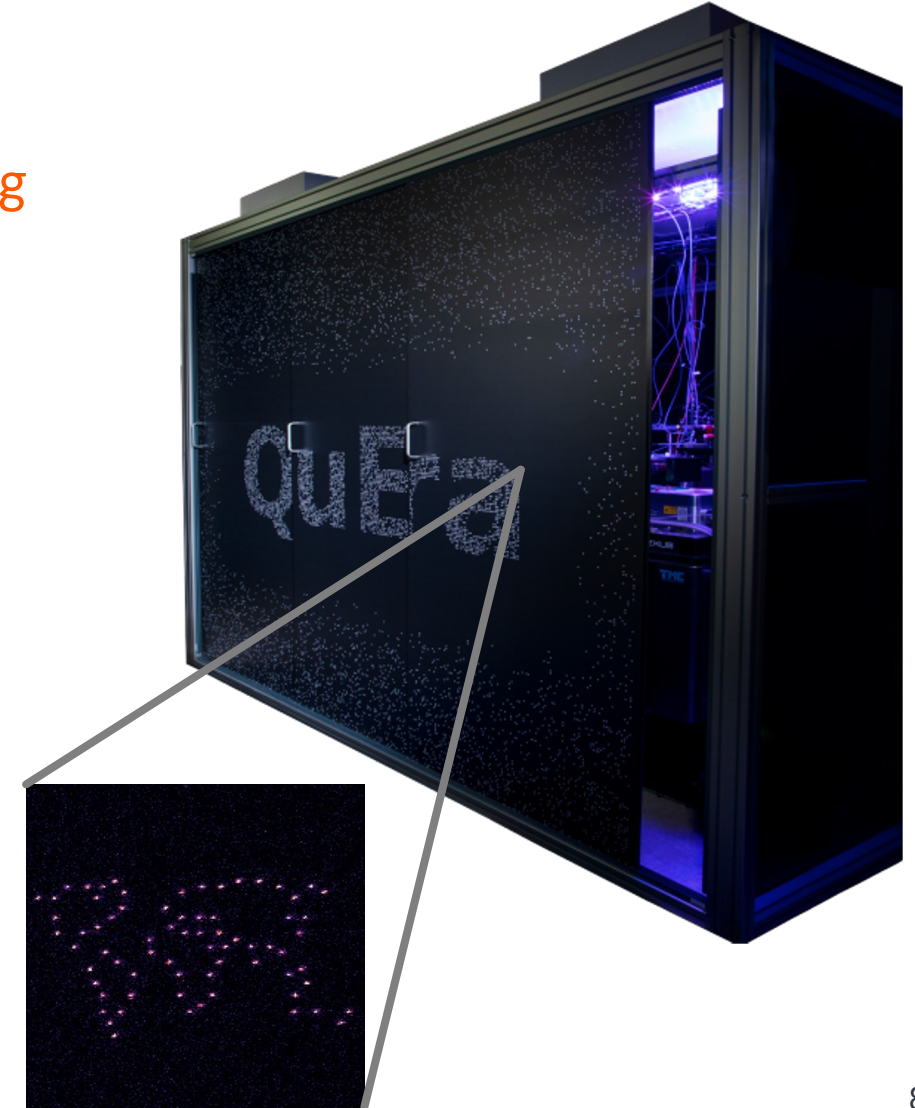
analog architecture

Aquila: QuEra's first-generation **analog** mode quantum computer

Implement programmable, nonequilibrium quantum dynamics on arrays of up to **256 qubits** with the **Rydberg Hamiltonian**

- "Science first" approach: **40 papers** have used Aquila since launch
- Publicly available on the cloud via **Amazon Braket** for over three years
- **130** hours/week, 99% uptime.

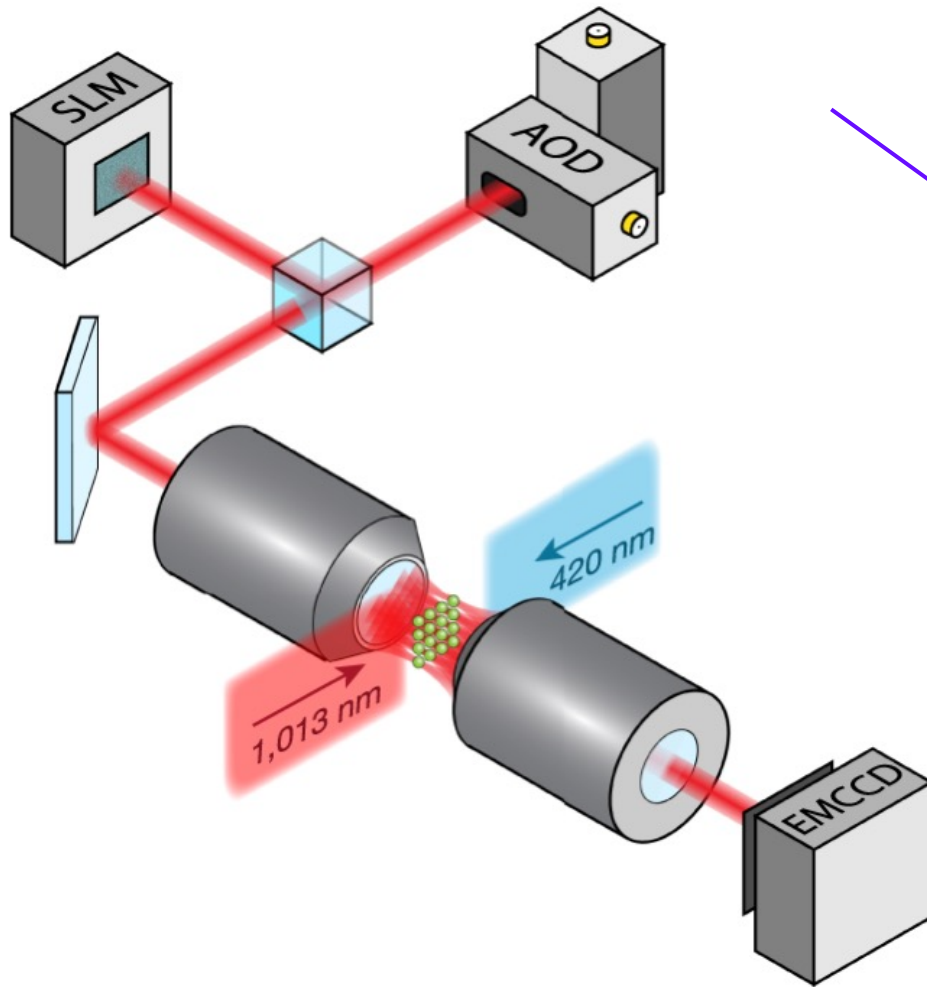
$$\begin{array}{c} |0\rangle \\ |0\rangle \\ |0\rangle \\ |0\rangle \\ |0\rangle \\ |0\rangle \end{array} \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \begin{array}{c} \mathcal{T} e^{-\frac{i}{\hbar} \int_0^t d\tau H(\tau)} \end{array} \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array}$$



Gemini: QuEra's first-generation **digital** mode quantum computer



Hardware elements overview

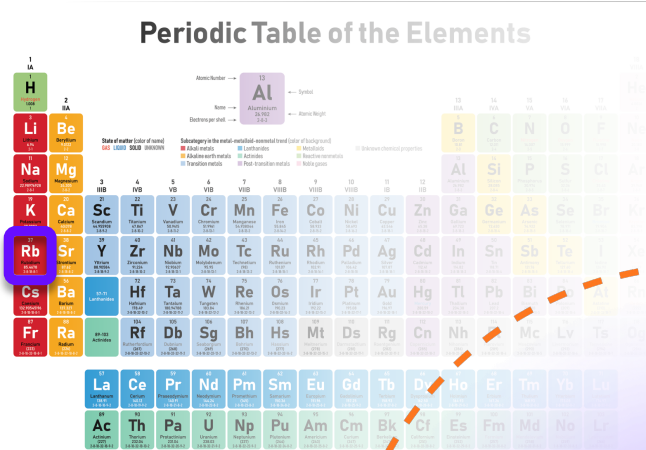


Gemini class

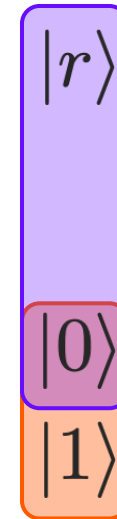
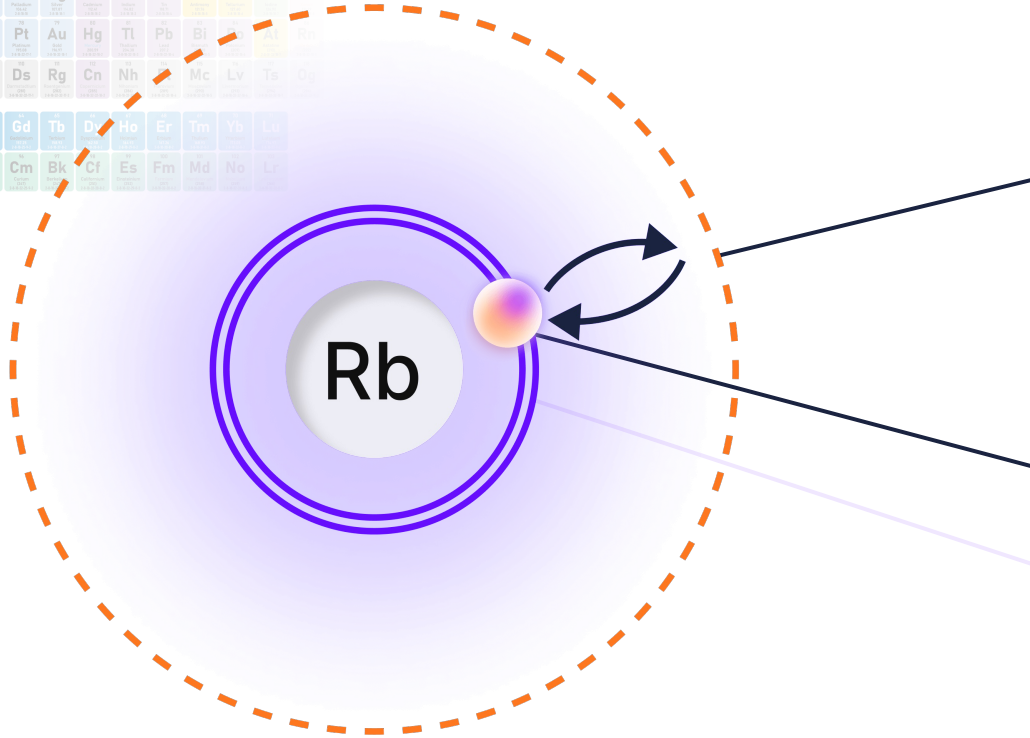


Rubidium qubit inner works

Periodic Table of the Elements

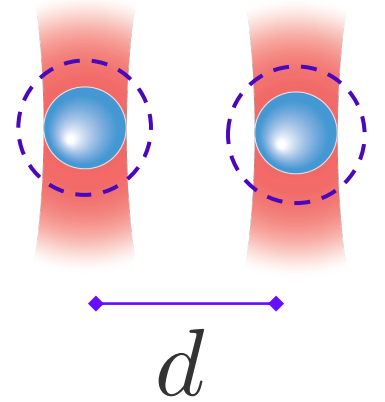


The periodic table shows elements from Hydrogen (H) to Oganesson (Og). Rubidium (Rb) is located in the first column, fourth row from the bottom. It is a soft, silvery-white metal that is highly reactive.



Entanglement through high-energy level

Strong geometrically
controlled interactions
 $V \sim d^{-6}$



Rydberg
state

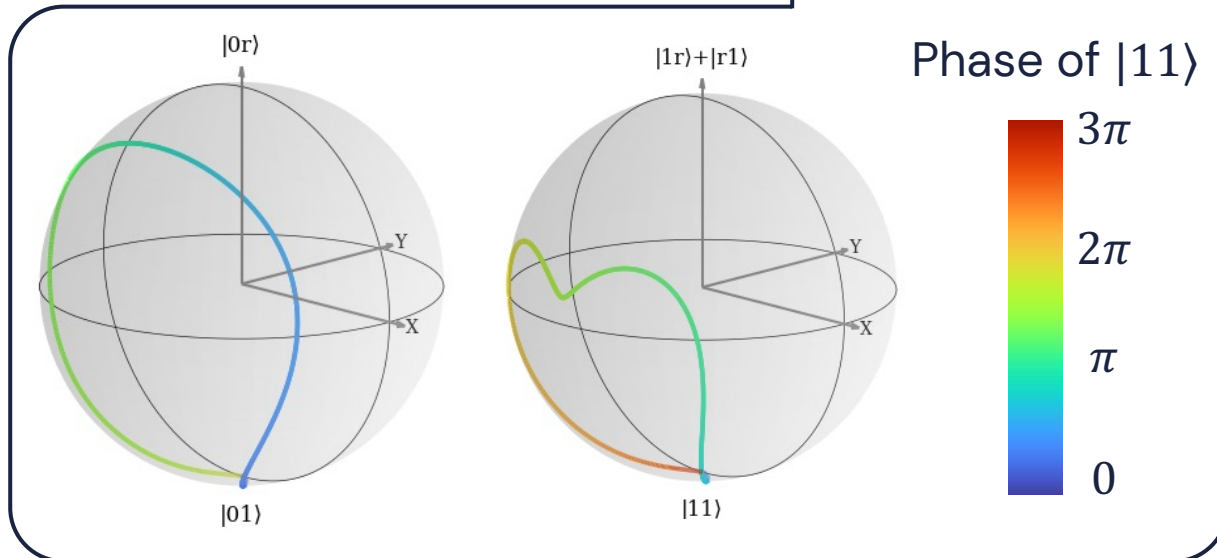
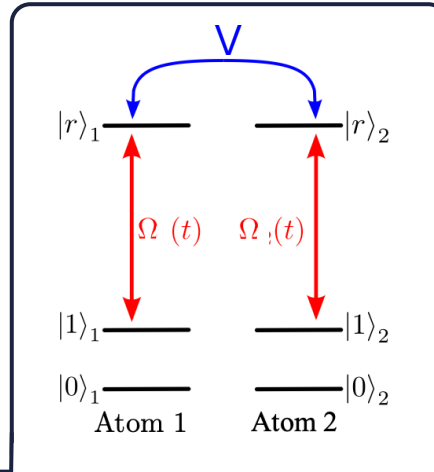
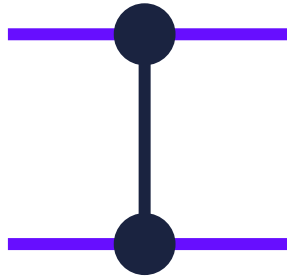
Hyperfine states

Long coherence times
High stability, flexibility

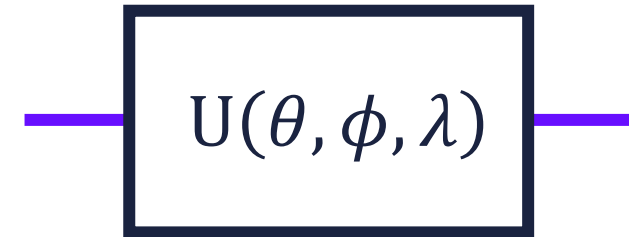
Information storage
in low-energy levels

Native gates

Controlled-Z gates



Arbitrary 1-qubit rotations (Z * XY-plane rotations)

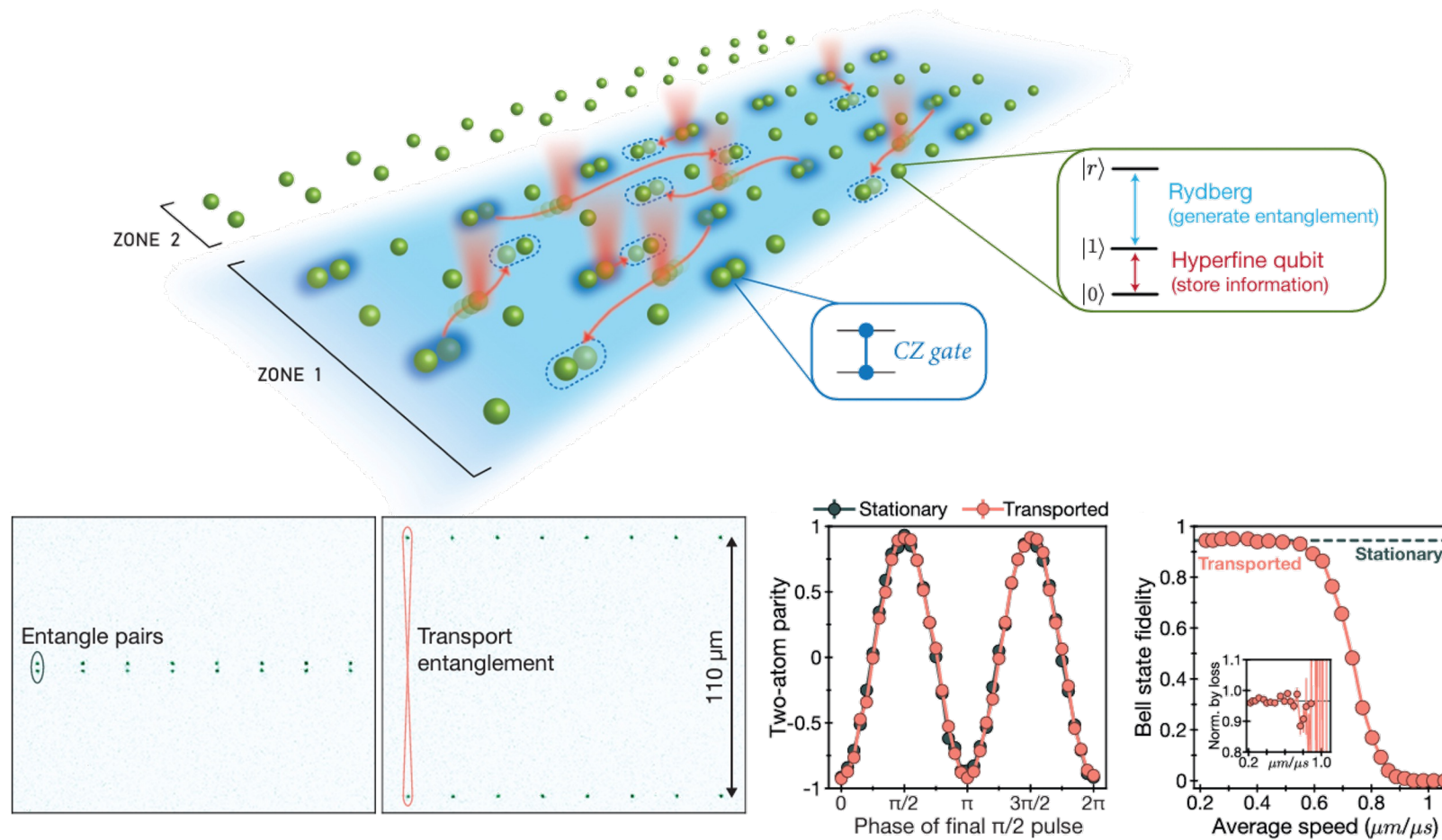


**Parallelism is inherent
to neutral atoms!**

cZ gates between each pair in a zone

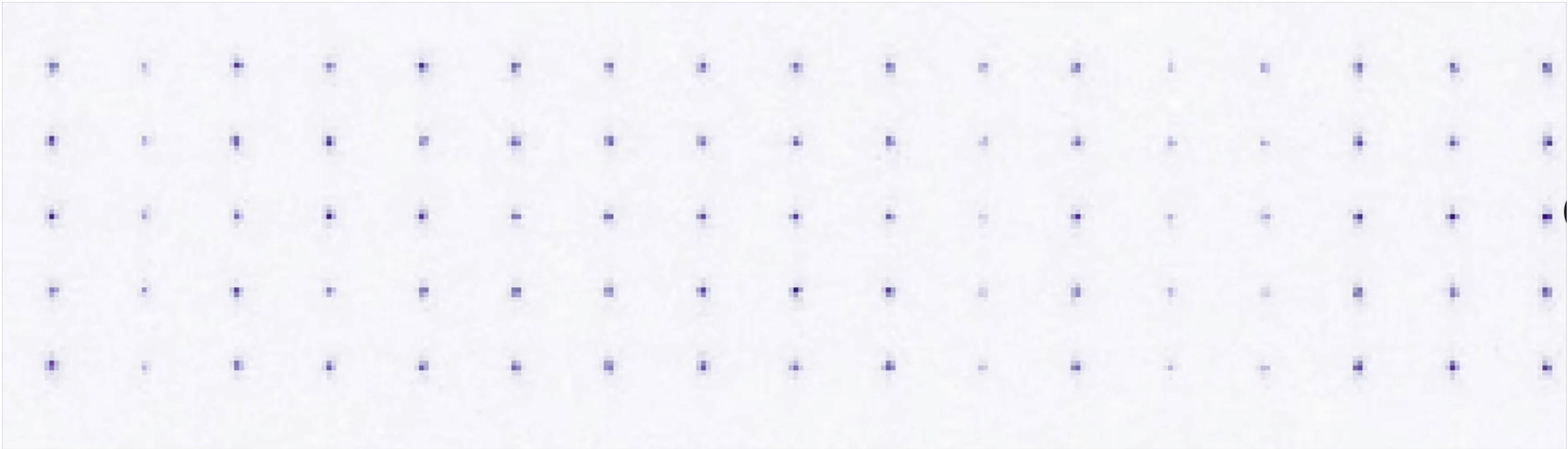
Global arbitrary 1 qubit rotations

Basic architecture: mid-circuit reconfigurability



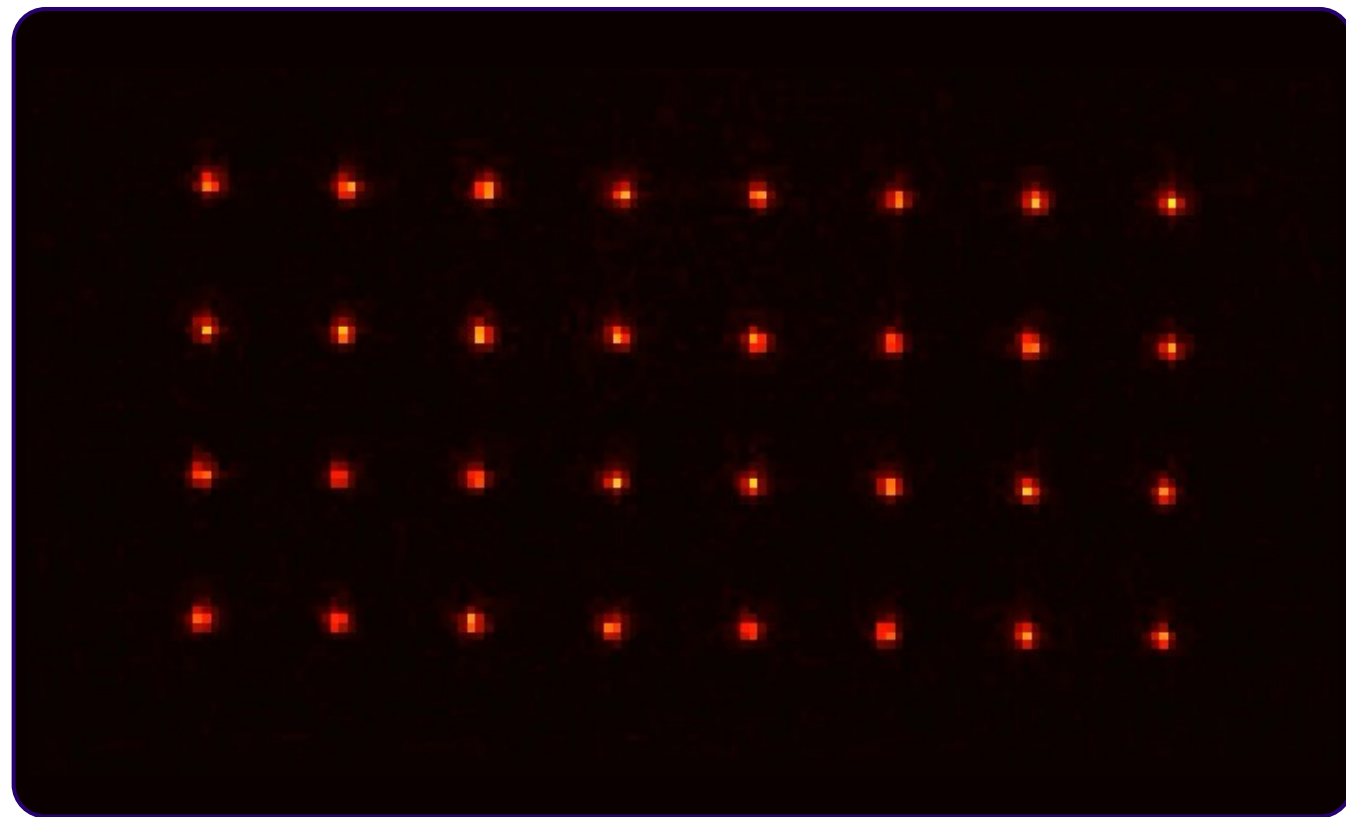
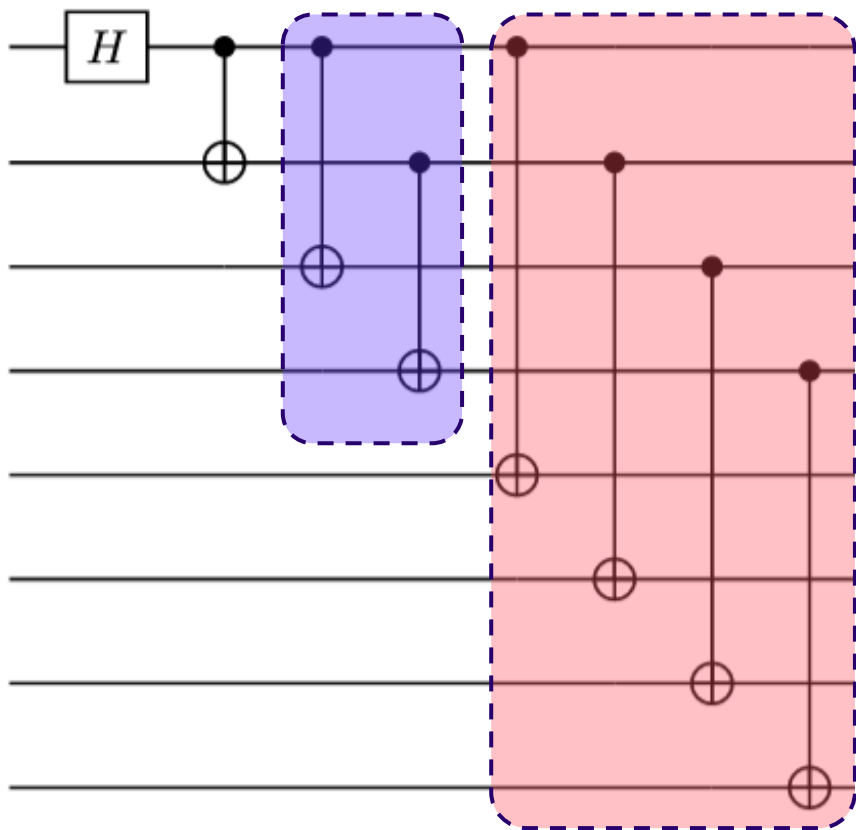
Bluvstein et al., Nature (2022)

Basic architecture: mid-circuit reconfigurability



[animated figure]

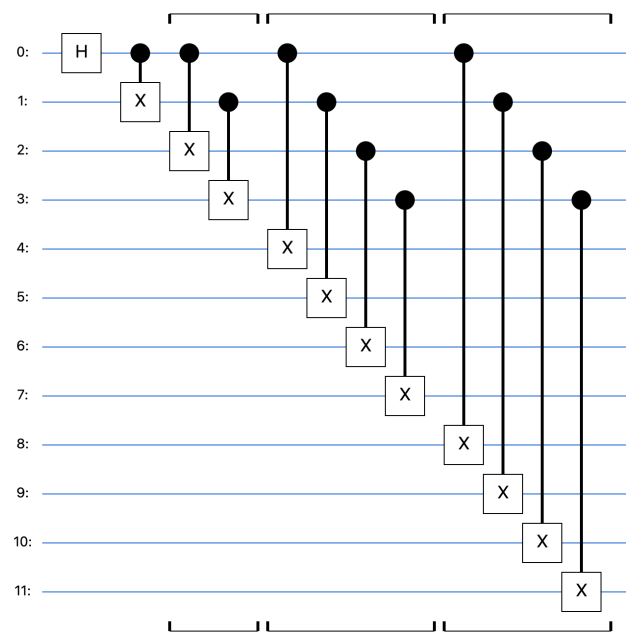
Gates + reconfiguration = quantum programs



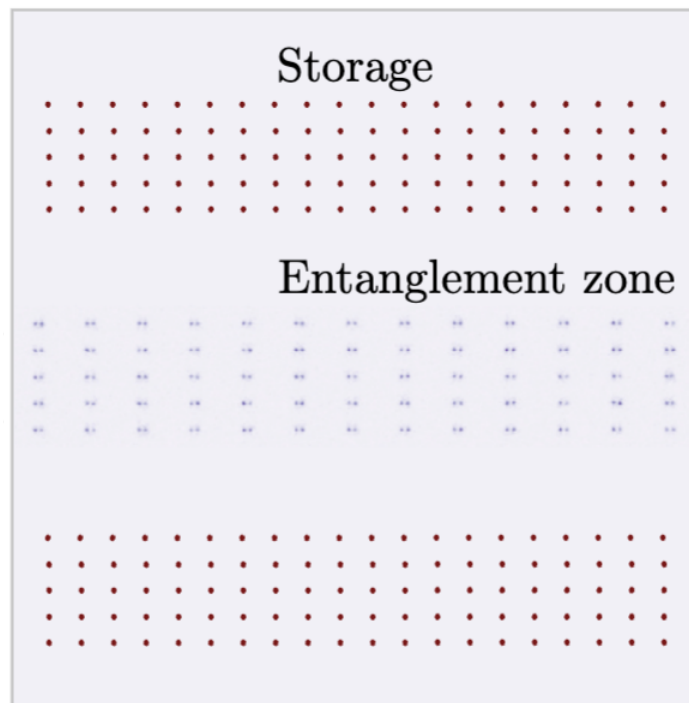
[animated figure]

Putting it all together

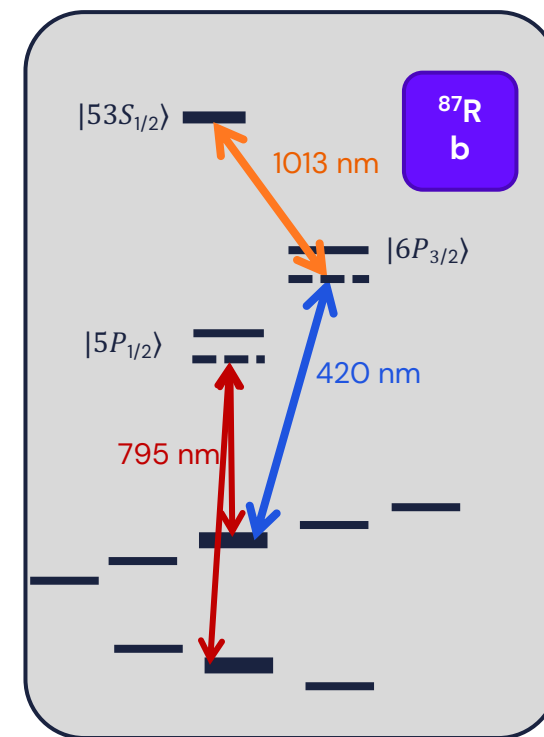
Gates / circuits



Moves



Pulses / AMO

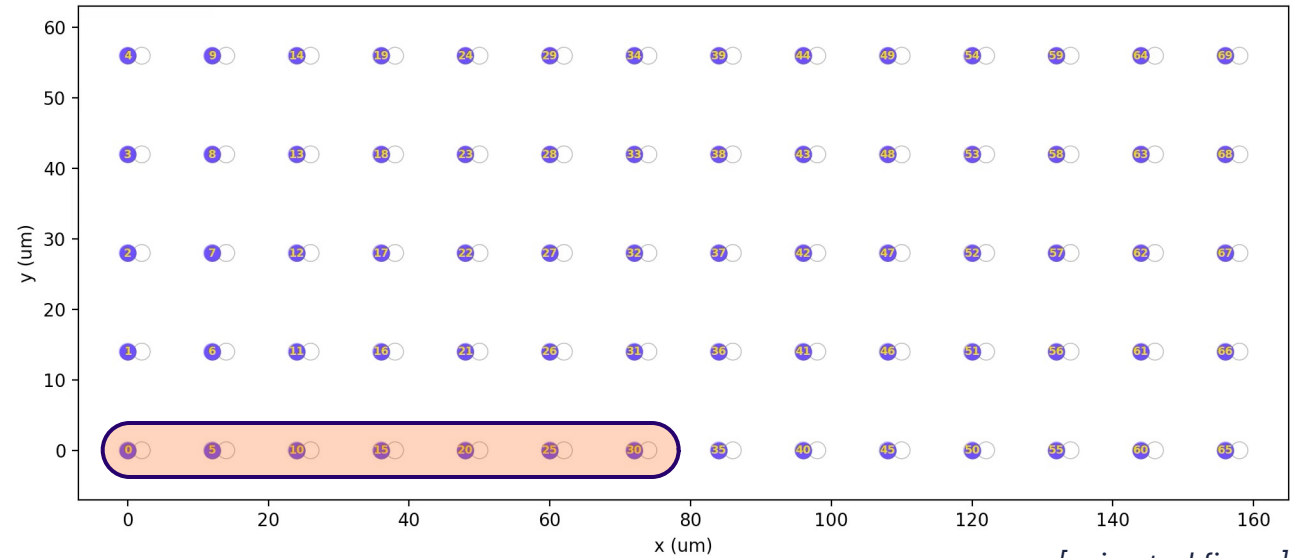


Compiler

Compiling circuits down to moves on Gemini:

```
@squid.kernel
def log_ghz_kernel():
    q = squid.galloc(n_qubits)
    squid.h(q[0])

    for level in range(max_iterations):
        width = 2**level
        for i in range(n_qubits):
            if i < width:
                target = i + width
                if target < n_qubits:
                    squid.cx(q[i], q[target])
    return q
```



[animated figure]

Logical qubit blocks with
transversal gates!

First, **prepare** Steane codes.

Then, **execute** circuit

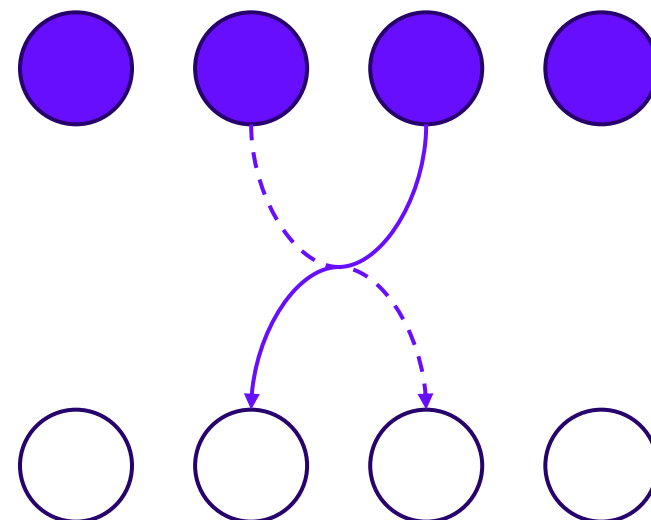
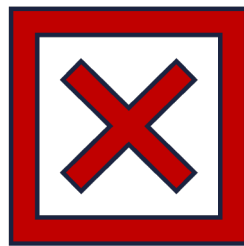
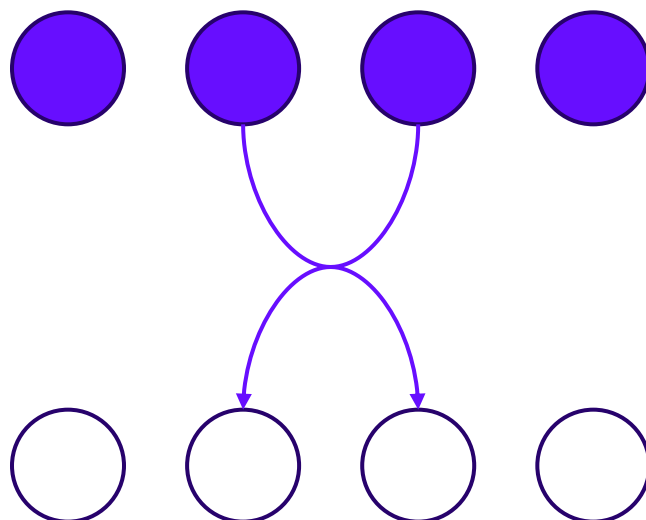
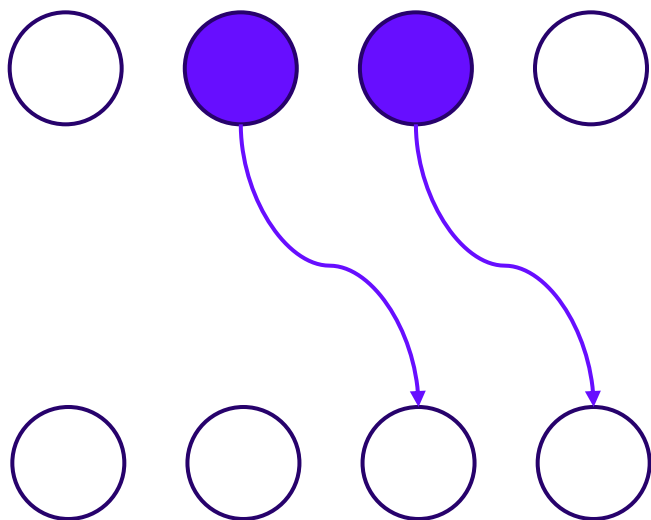
How do we program Gemini?

What are good (and bad) **programming patterns** for reconfigurable atom arrays?

What is the **noise model** of neutral atoms?

How do we **define** quantum executions?

Atom shuttling rules #1 – crossing conflict



Atom shuttling rules #1 – crossing conflict



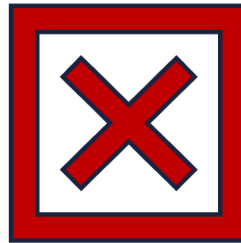
[animated figure]



[animated figure]



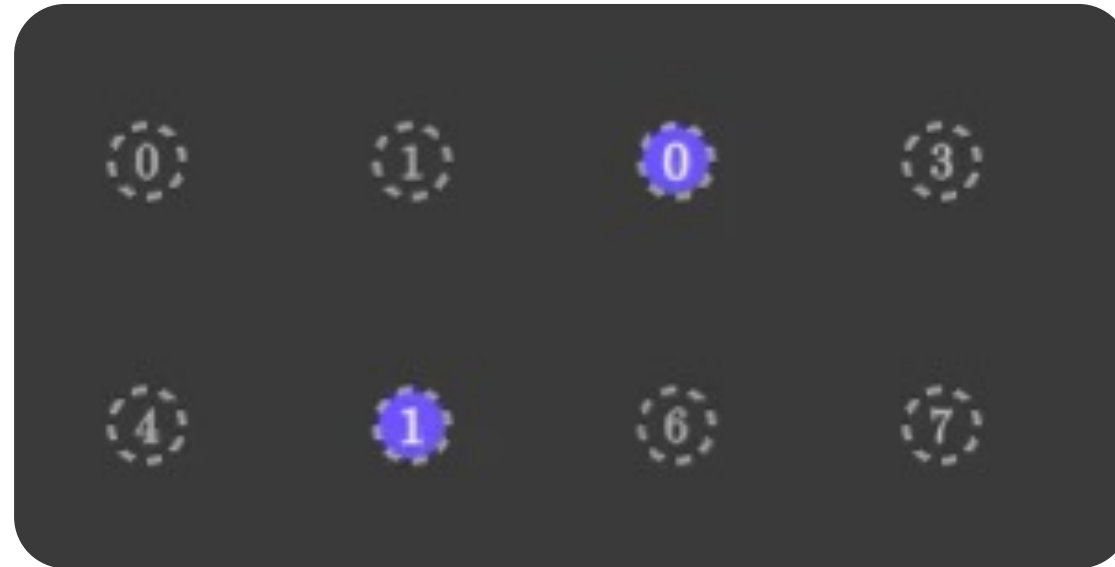
[animated figure]



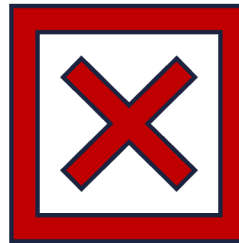
“atoms cannot collide”

“atoms cannot change order in a single move”

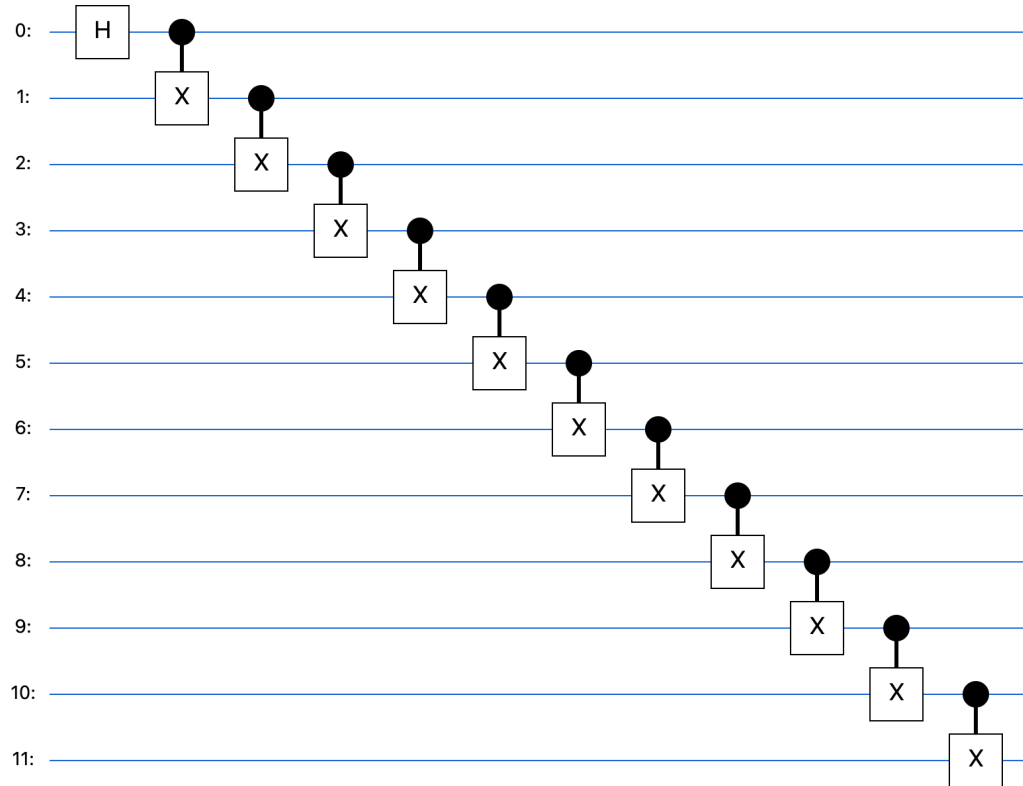
Atom shuttling rules #2 – “many-to-one” conflict (bonus)



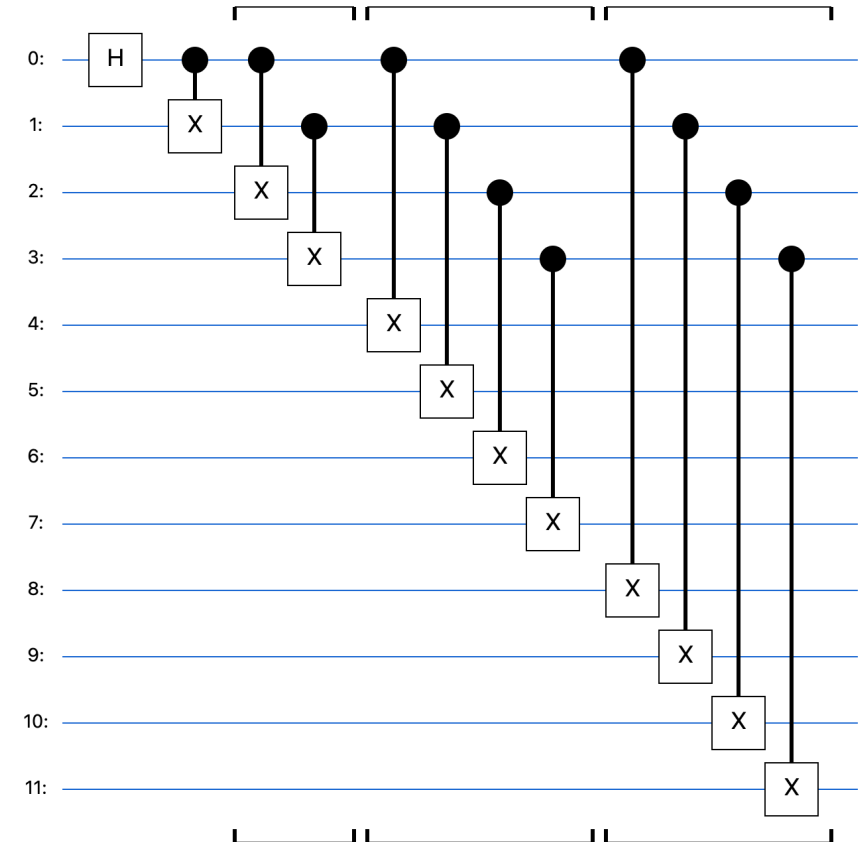
[animated figure]



Mid-circuit reconfiguration means long-range connectivity

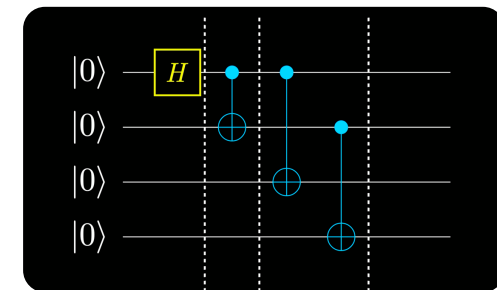


Linear-depth GHZ state preparation
using nearest-neighbor connectivity

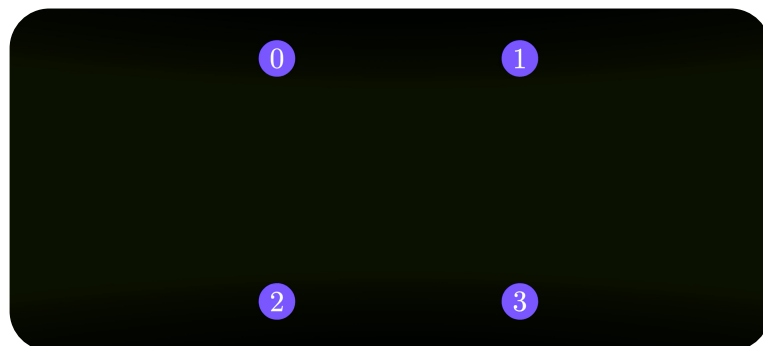
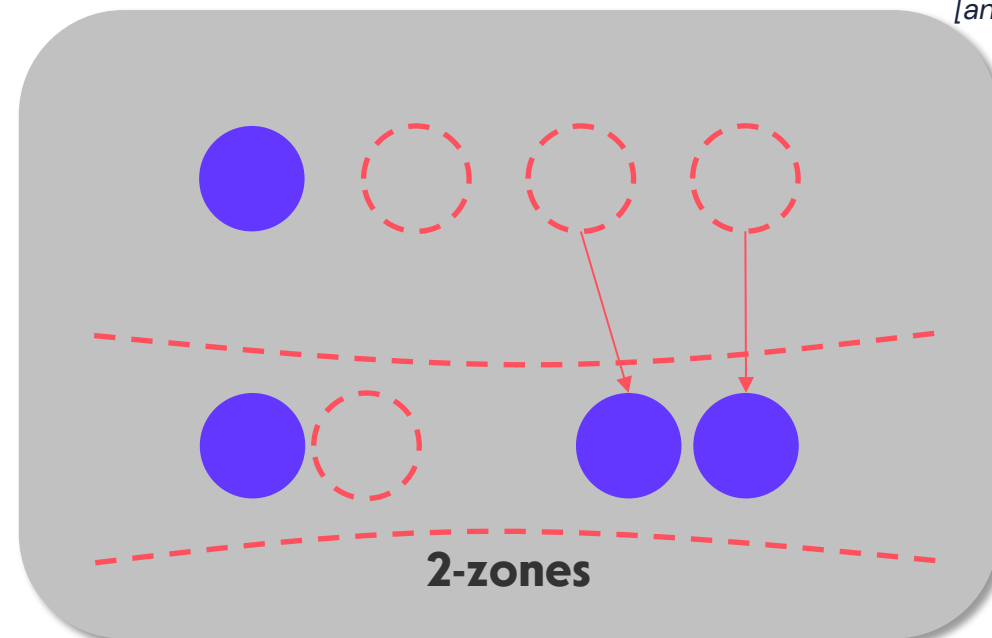
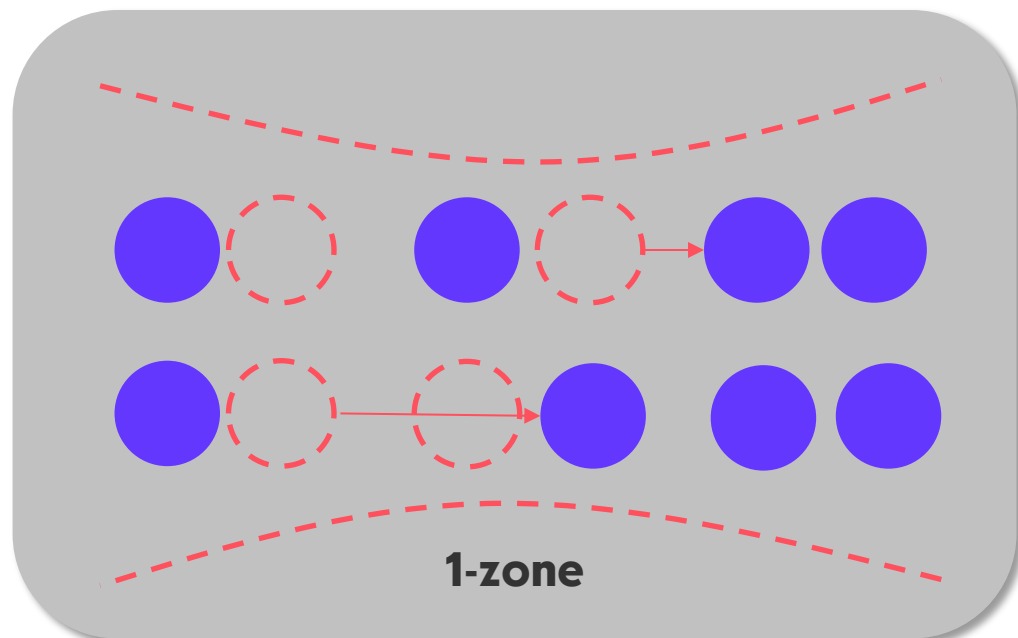


Log-depth GHZ state preparation
using arbitrary connectivity.
Minimize the number of sequential layers!

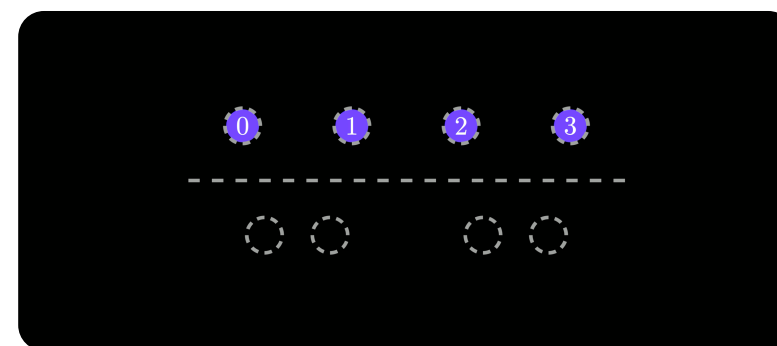
Zoned architectures



[animated figure]

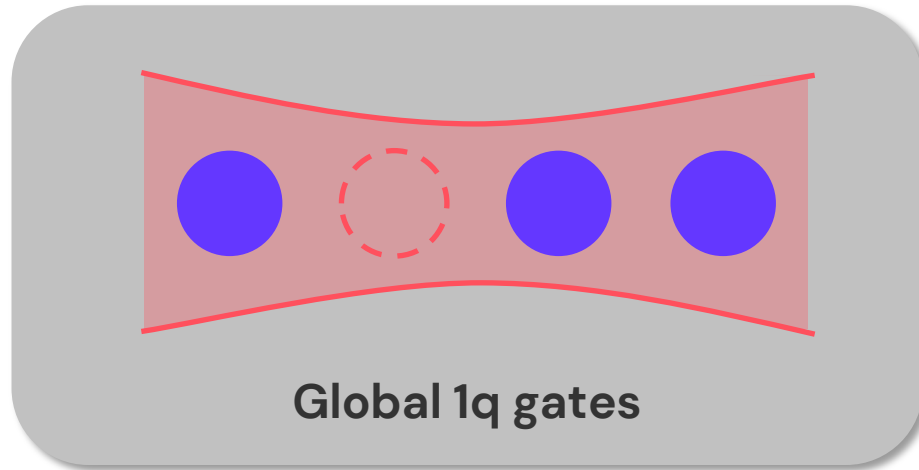


[animated figure]

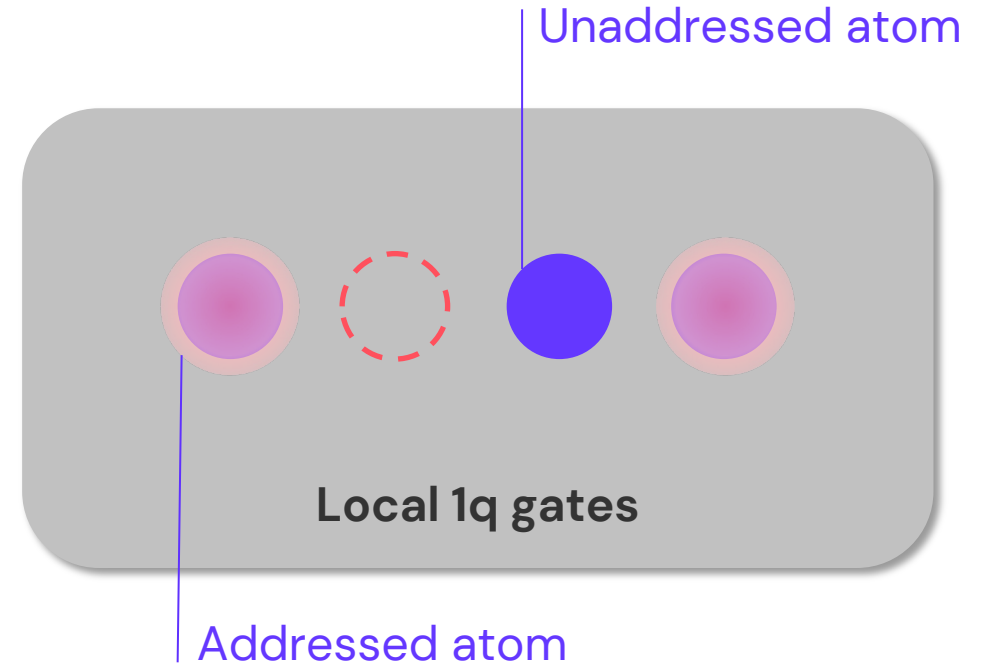


[animated figure]

Error channels – single qubit operations

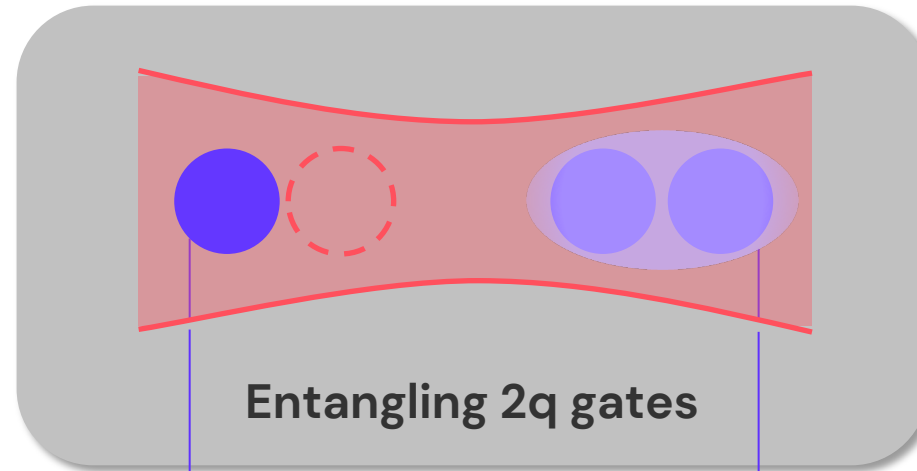


Error budget $\sim 10^{-5}$



Error budget $\sim 10^{-4}$

Error channels – two-qubit operations



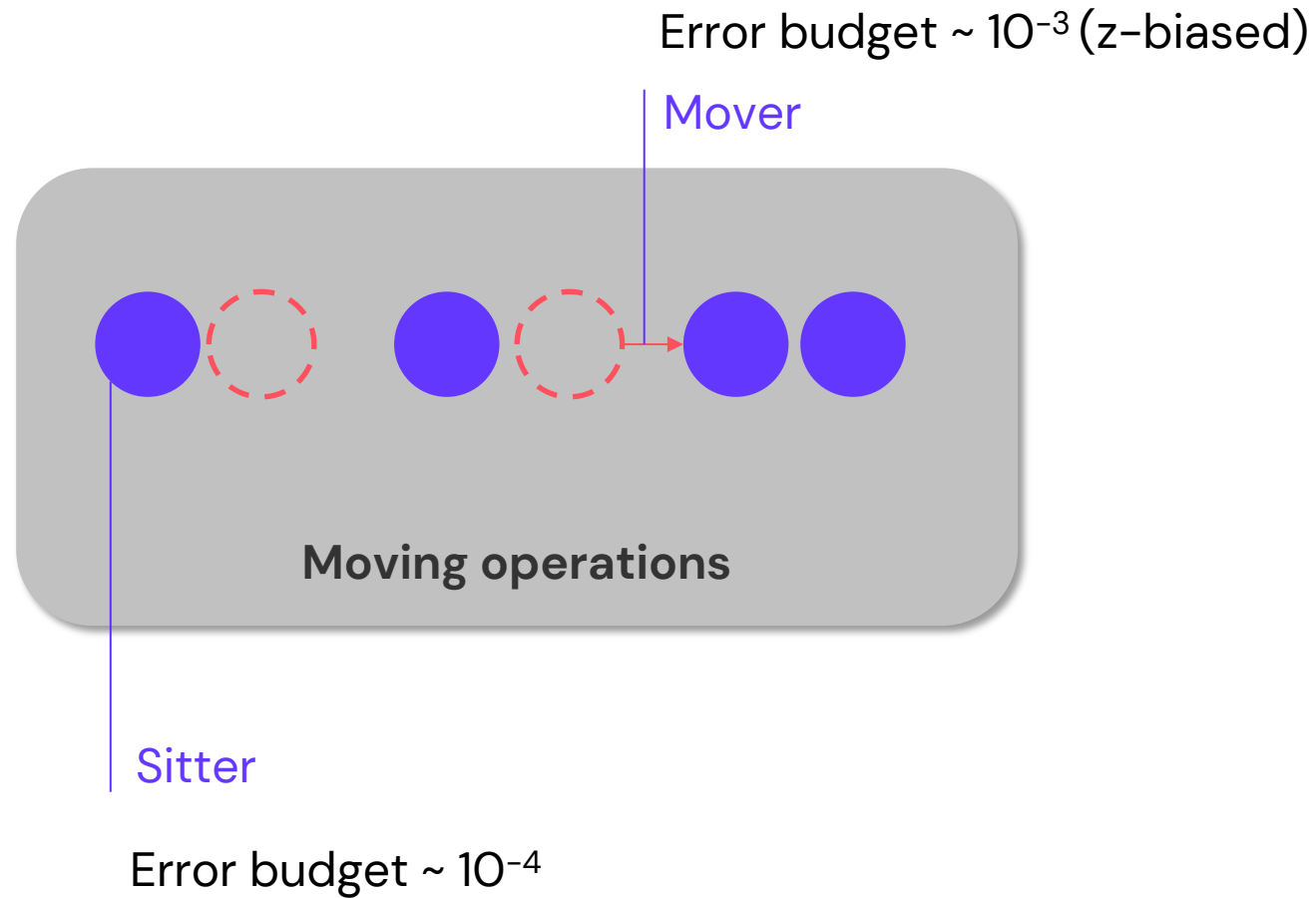
Unpaired/spectator atom

Error budget $\sim 10^{-3}$ (z-biased)

Paired/entangled atoms

Error budget $\sim 10^{-3}$ (z-biased)

Error channels – shuttling

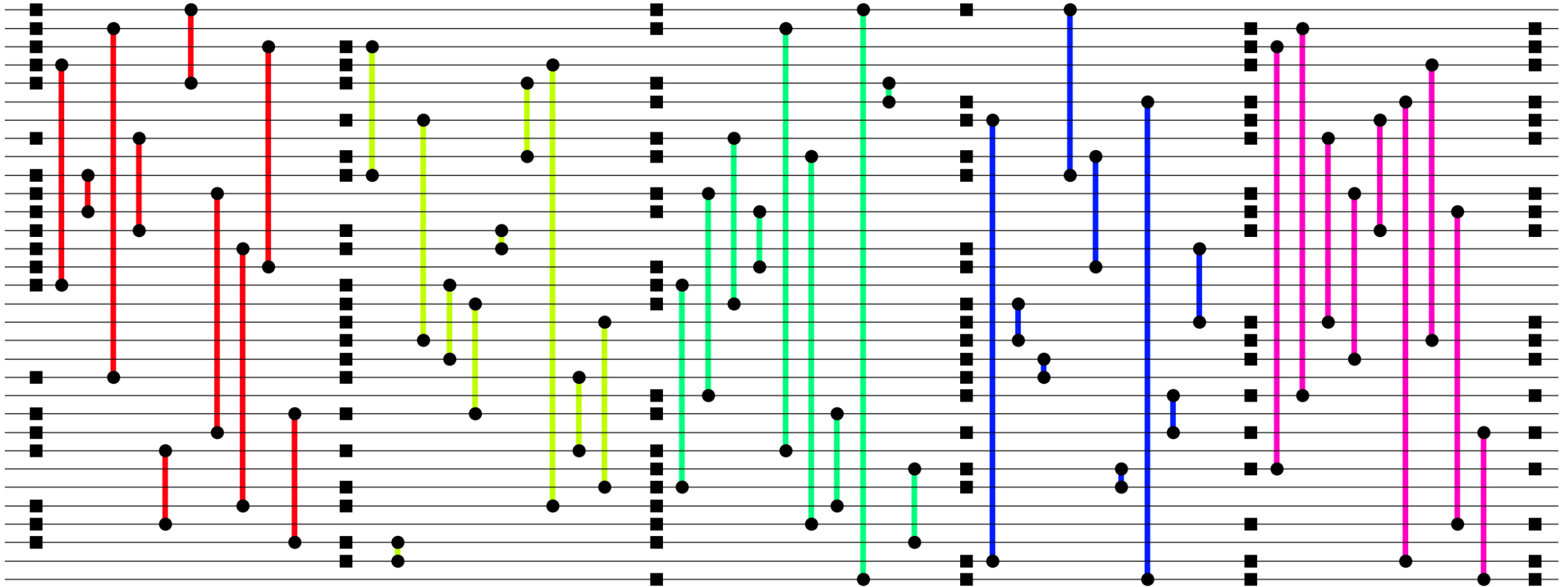


Noise hierarchy

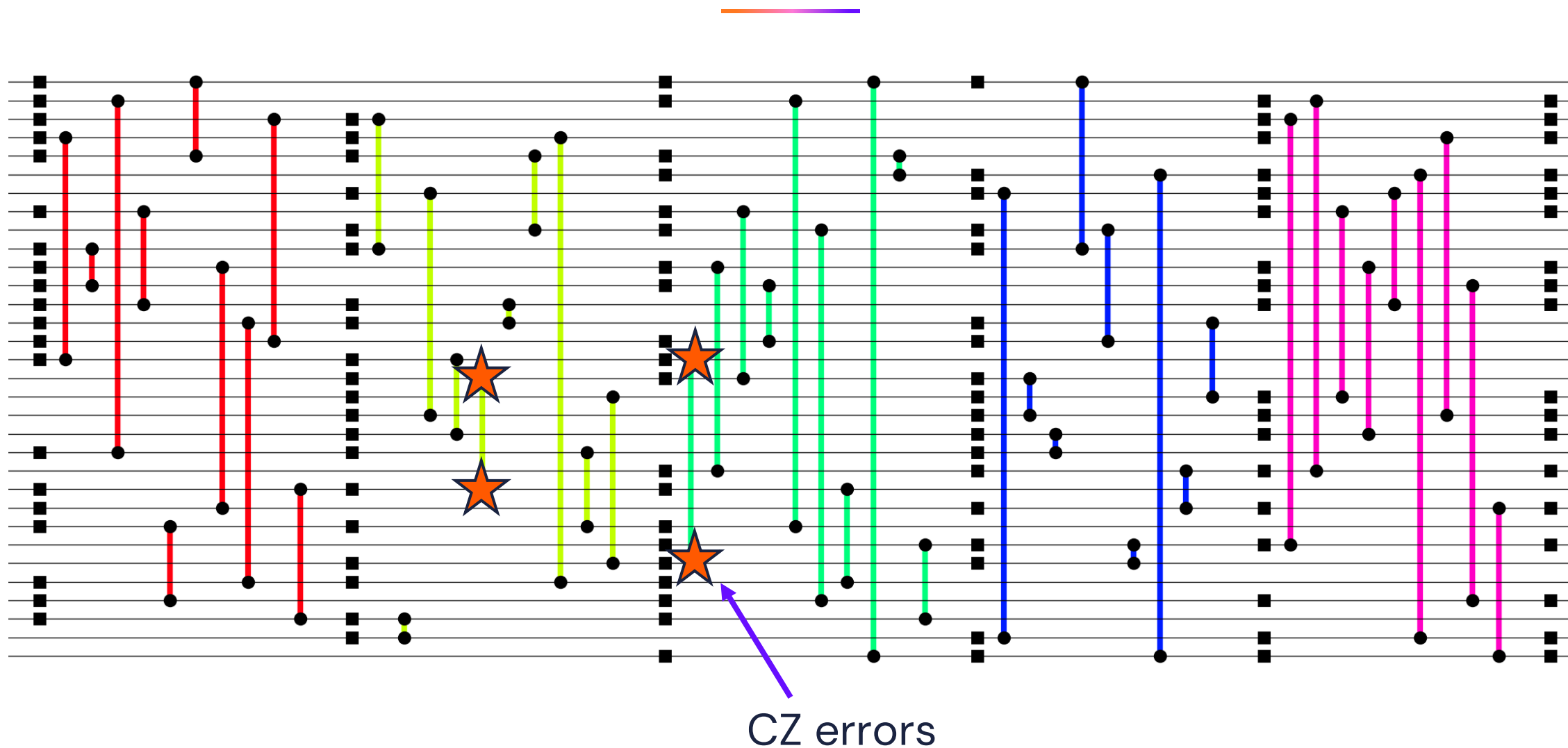
$$E_{CZ} \gtrsim E_{mover} \sim E_{unpaired} > E_{sitter} \sim E_{1-qubit,local} > E_{1-qubit,global}$$

Z-biased

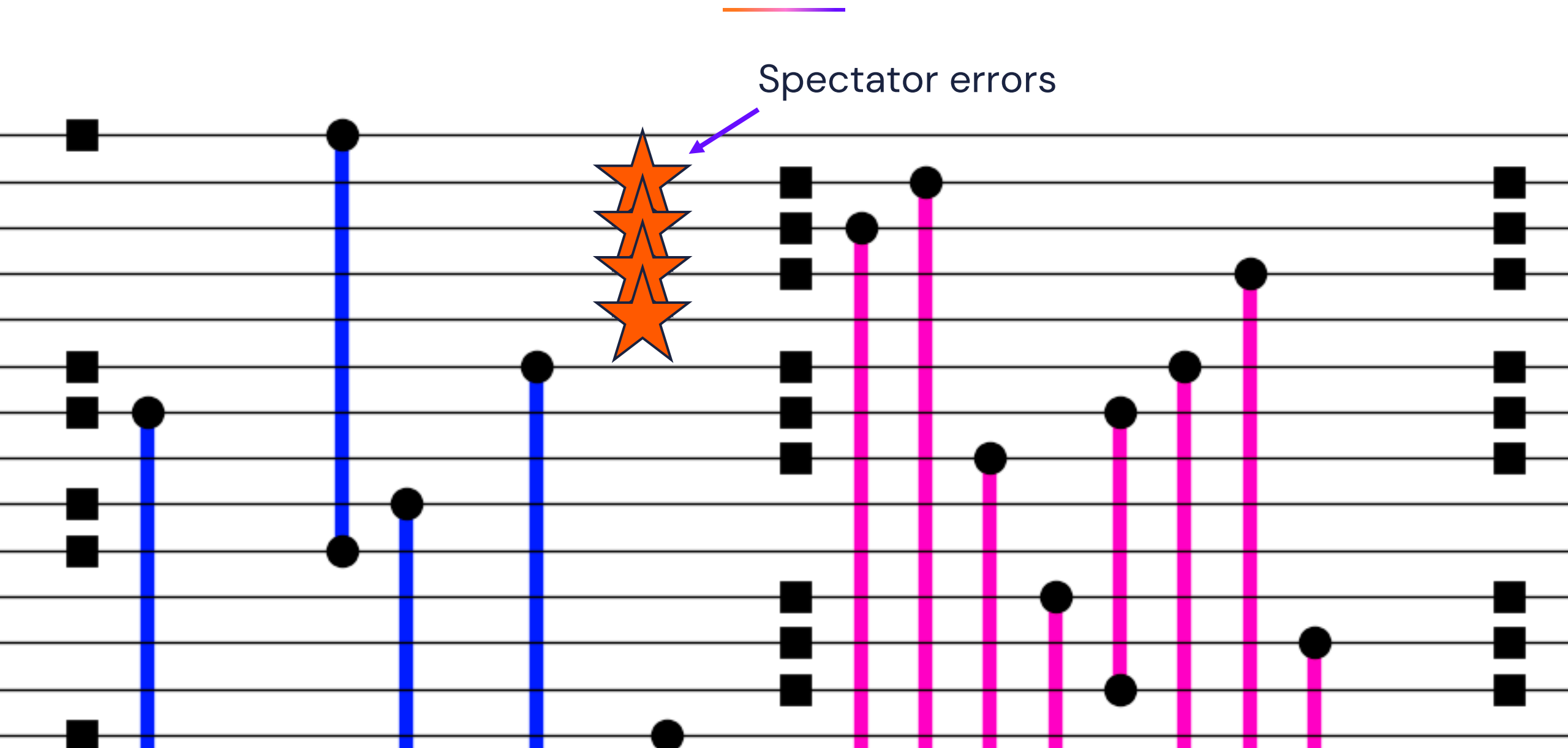
Program design goals



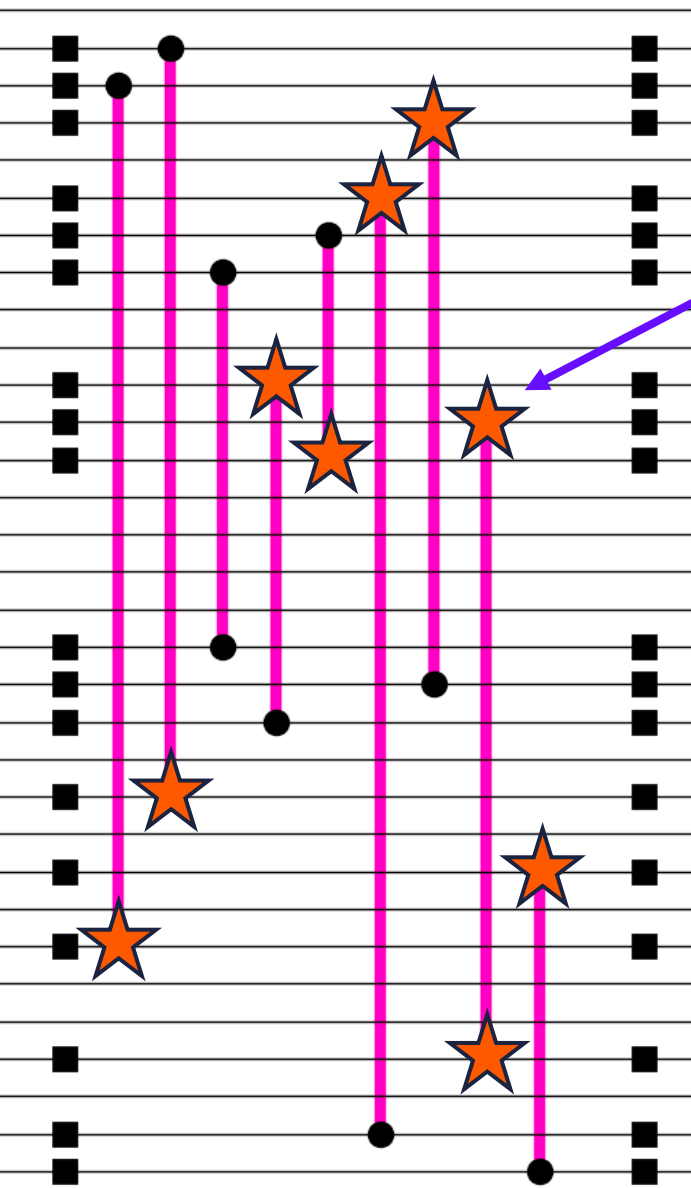
Heuristic: Minimize number of CZ gates



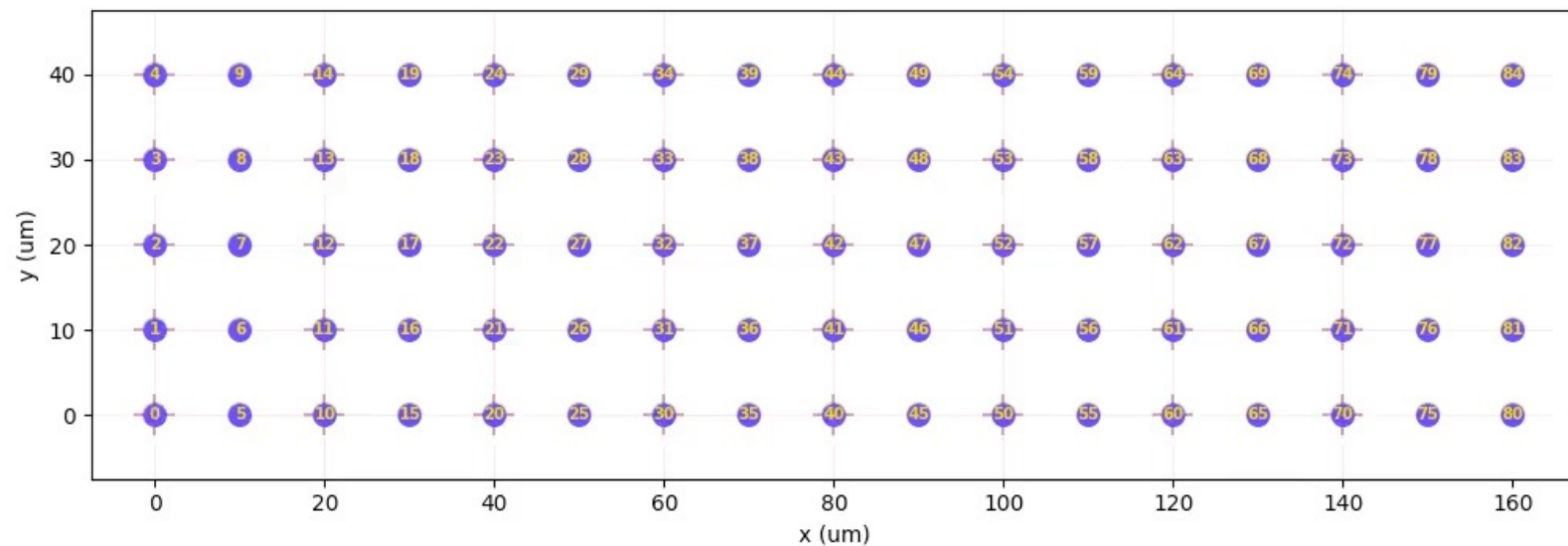
Heuristic: Maximize parallelism



Heuristic: Fit hardware connectivity

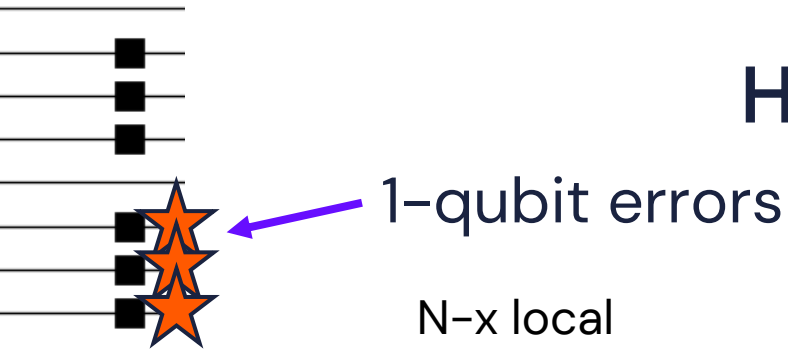


Move errors

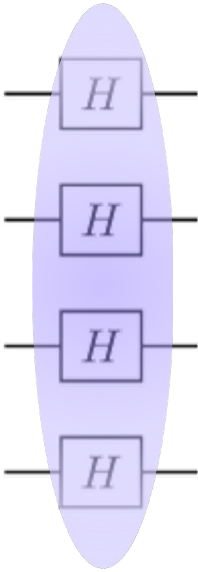


[animated figure]

Heuristic: Global vs local gates

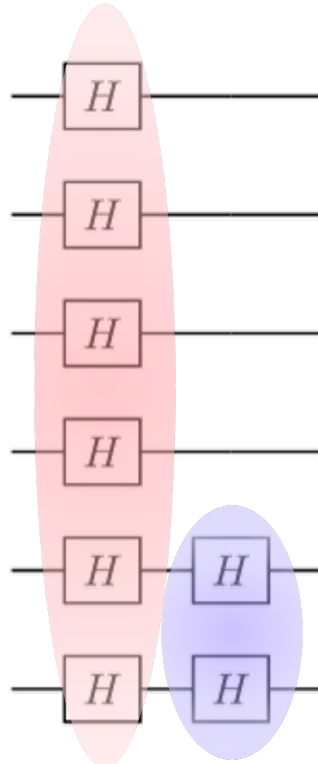


N-x local

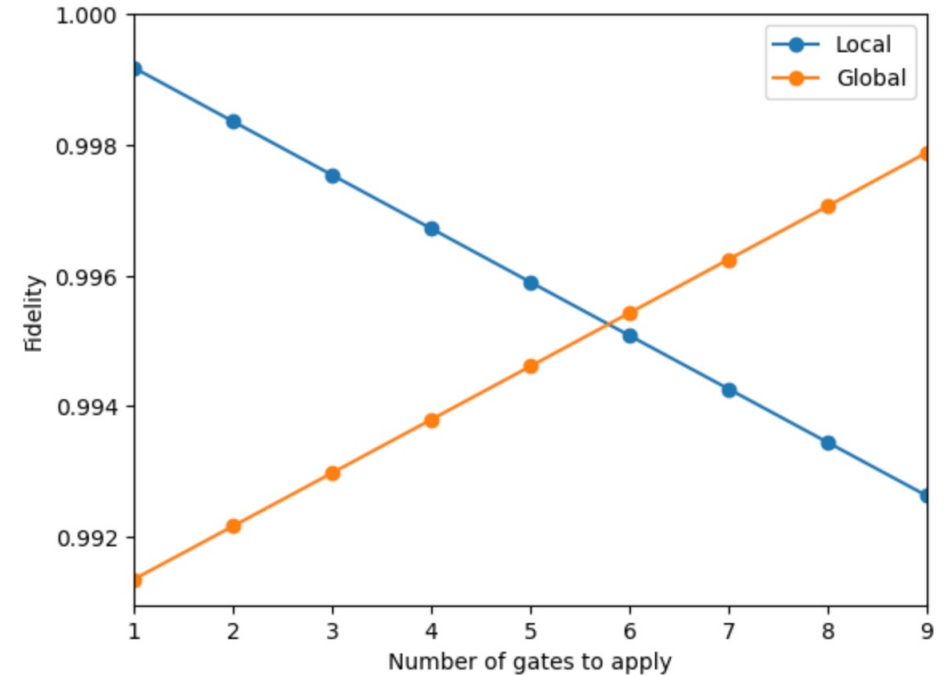


VS

1 global



x local



$$F = (1 - \epsilon_L)^{N-x}$$

$$F = (1 - \epsilon_G)^N (1 - \epsilon_L)^x$$

$$x_c \approx 0.421N$$

Lesson:

If a circuit has a moment with more than 60% equal 1-qubit gates, it is better to exchange for a global and undo the undesired gates

-- 15 minute break --

In Part 2: how to use Bloqade to write programs compatible with Gemini

Follow along by cloning the following repository:

https://github.com/QuEraComputing/quera-education/blob/main/NEDO_2025/nedo_tutorial.py

How to program and analyze Gemini using Bloqade

Bloqade

Bloqade
Analog

Analog
Hamiltonian
Simulation



QuEra
Analog mode
Hardware (Aquila)

Bloqade
Circuit

Cirq suite

Noise modeling

Squin

Circuit synthesis



Bloqade
Shuttle

Atom shuttle
move scheduling



built with:

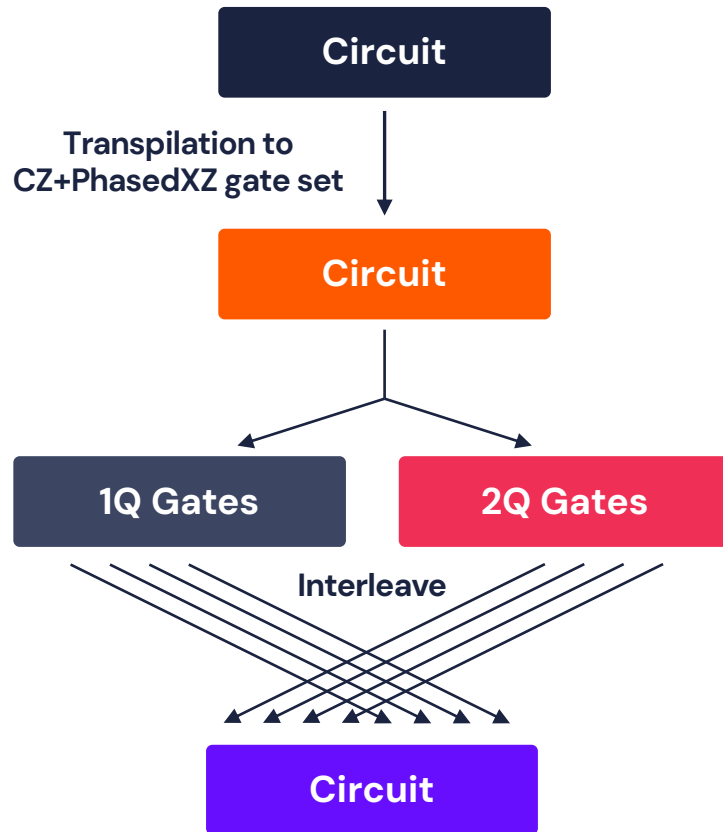


Kirin

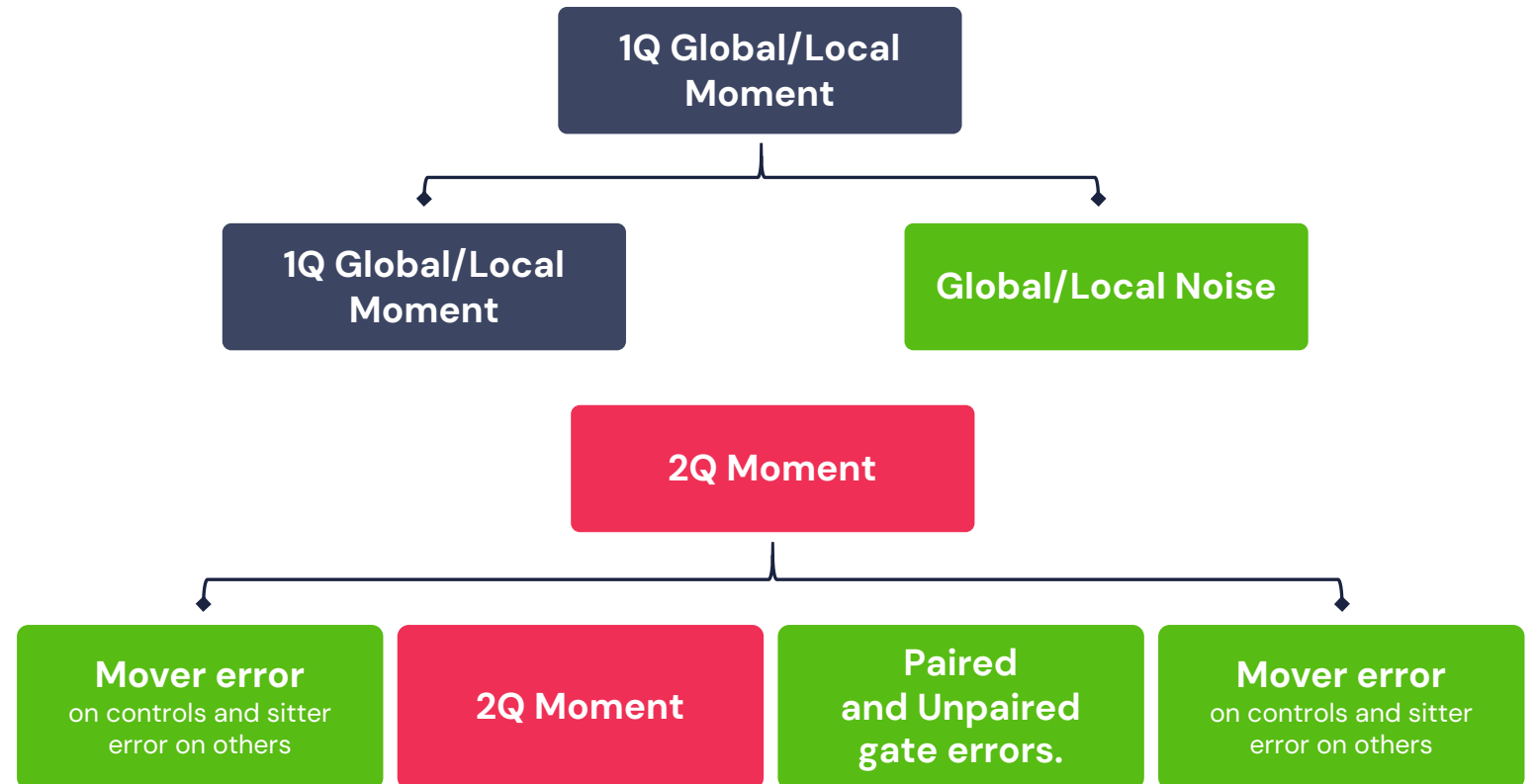
Compiler toolchain

Heuristic noise model – one-zone logic

Step 1

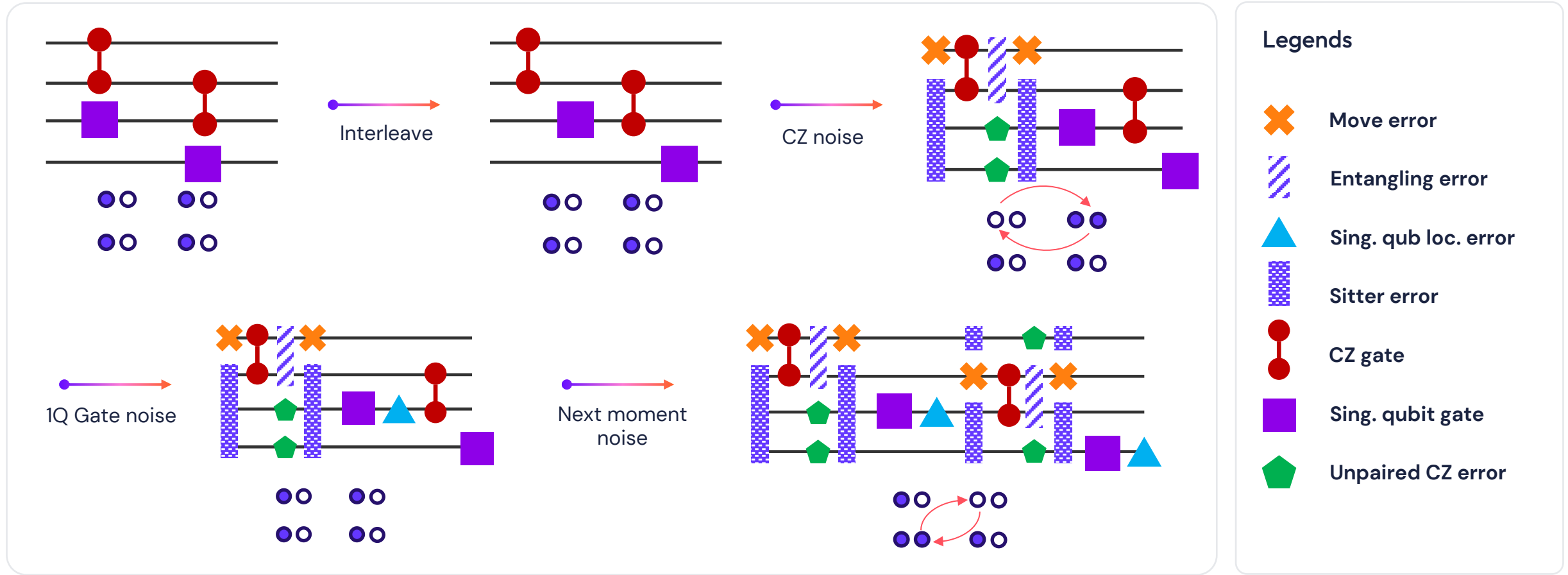


Step 2



Heuristic noise model - one-zone logic

Example



Let's explore some code!



https://github.com/QuEraComputing/quera-education/blob/main/NEDO_2025/nedo_tutorial.py

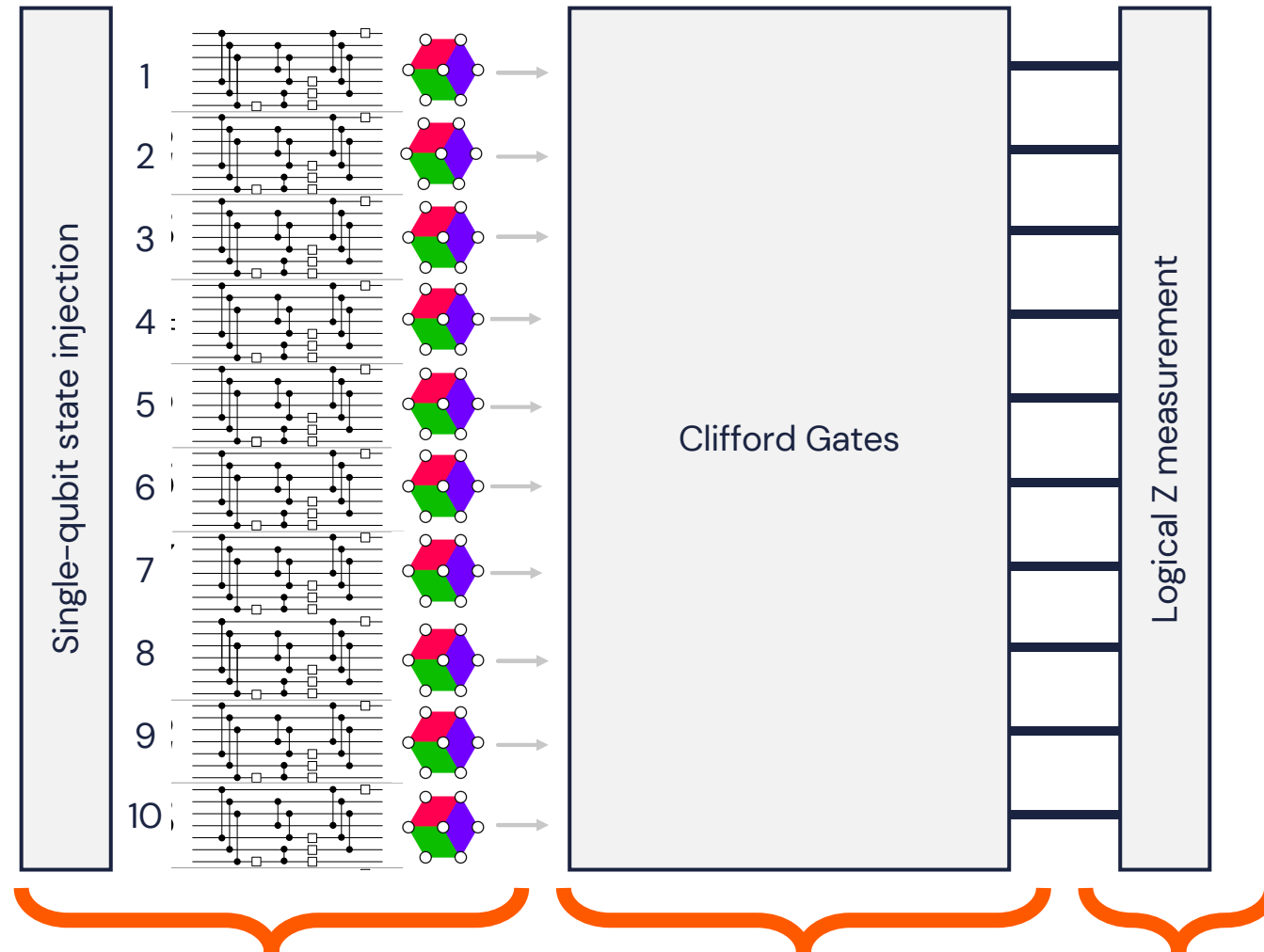
Other resources:

https://bloqade.quera.com/latest/digital/tutorials/circuits_with_bloqade/
https://bloqade.quera.com/latest/digital/examples/interop/noisy_ghz/

Bonus: Logical mode Gemini



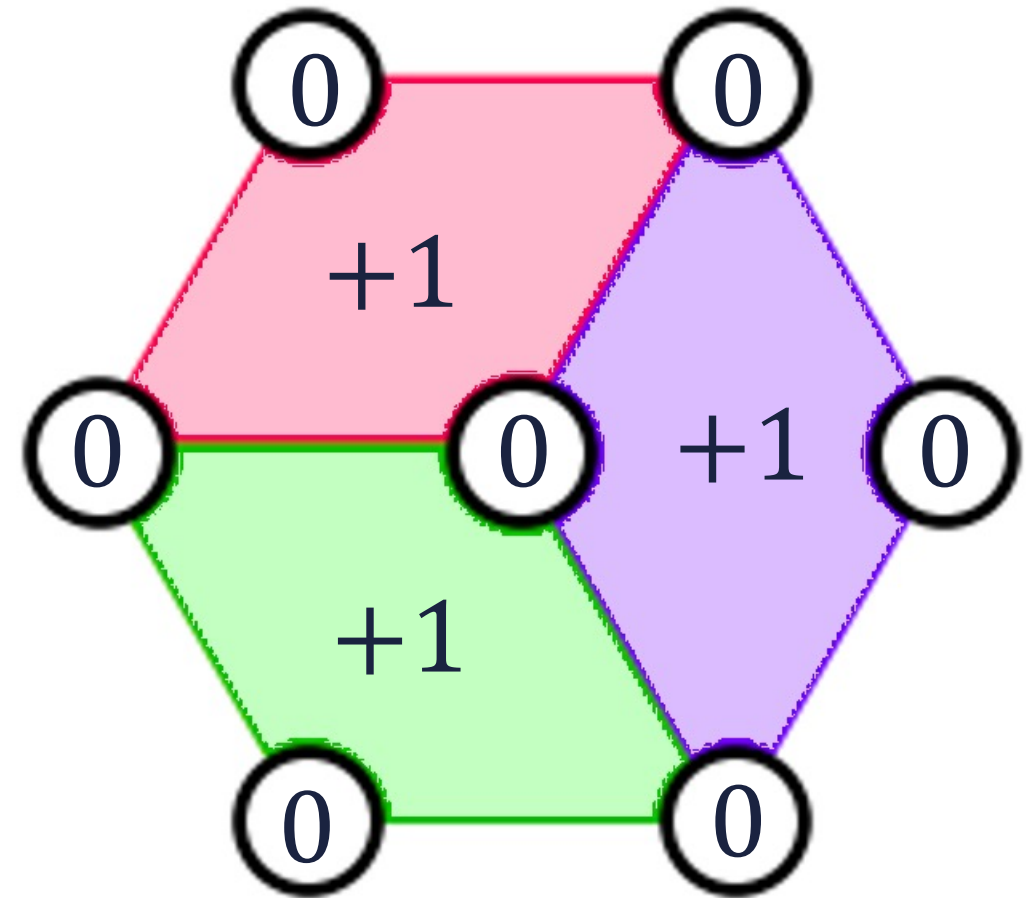
Logical mode Gemini: the initial mode for hardware access



A crash course in error correction

Error correction: extract errors from syndrome parity measurements

Color code: syndromes are 4-bit parity measurements. **7 bits can correct 1 error!**

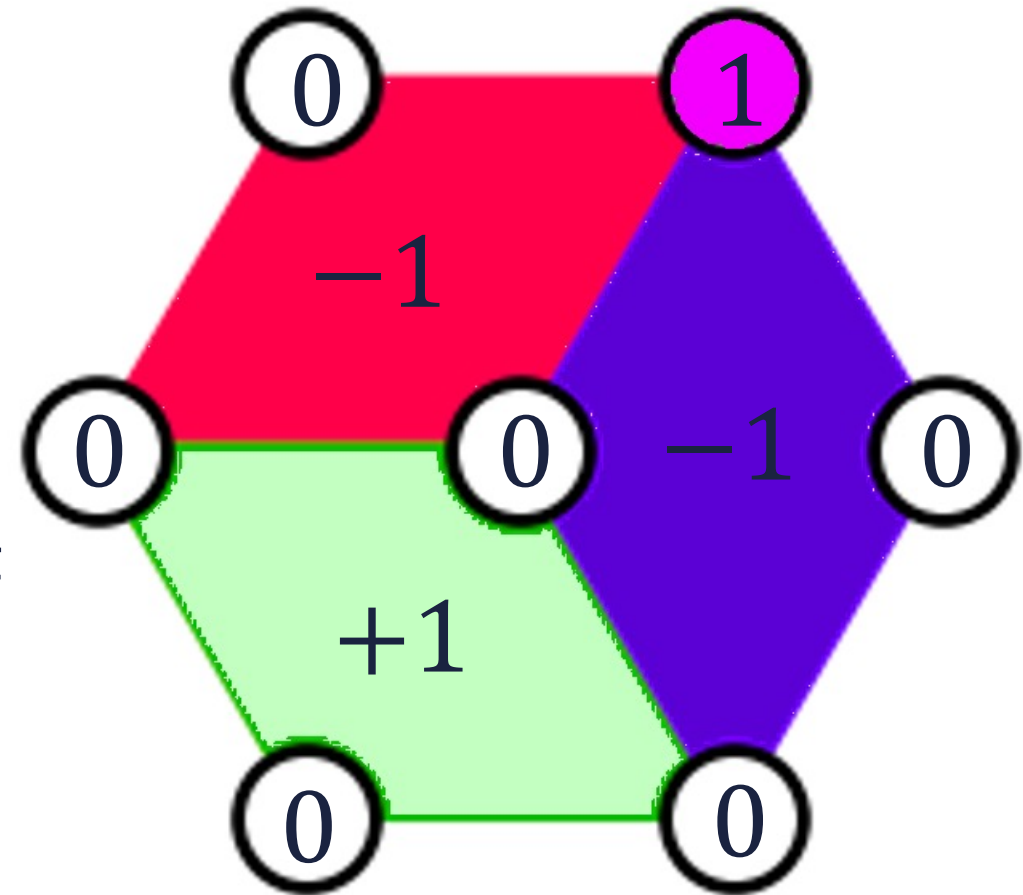


A crash course in error correction

Error correction: extract errors from syndrome parity measurements

Color code: syndromes are 4-bit parity measurements. **7 bits can correct 1 error!**

Quantum error correction: correct both X (bit flip) and Z (phase flip) errors without destroying the encoded qubit. Syndromes are stabilizers XXXX and ZZZZ



Gemini is a **testbed** for QEC

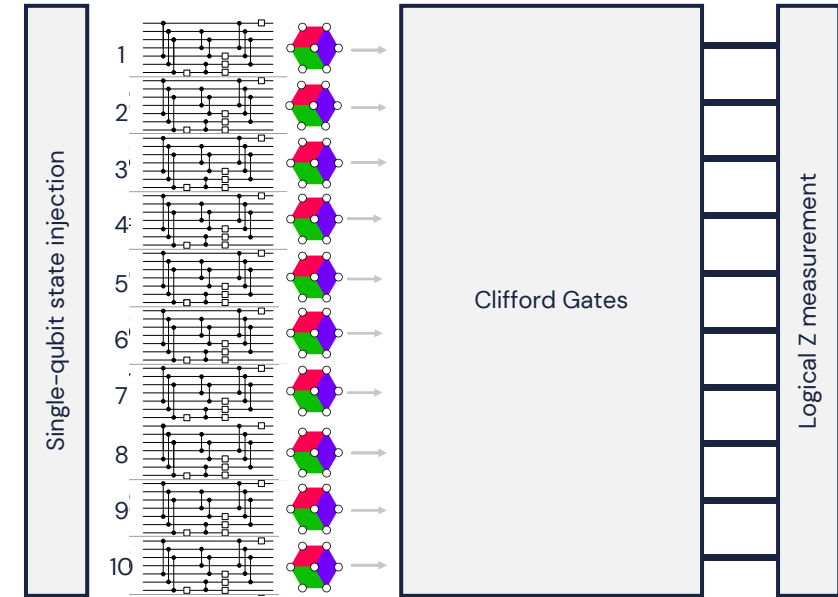
10 logical qubits $\Rightarrow$ Not enough to do anything “utility scale” or “nontrivial”

70 physical qubits $\Rightarrow$ Very difficult to simulate classically. **70 physical qubits with low magic** $\Rightarrow$ Interesting QEC explorations and demonstrations.

Recreate QuEra’s logical magic state distillation work

Develop decoder strategies and noise analysis

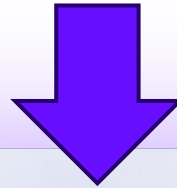
Demonstrate small QEC primitives



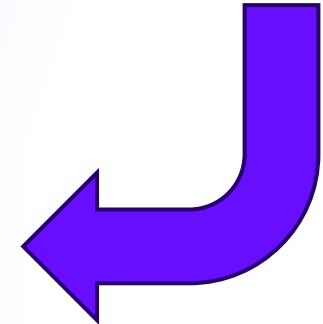
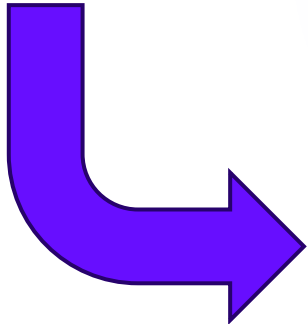
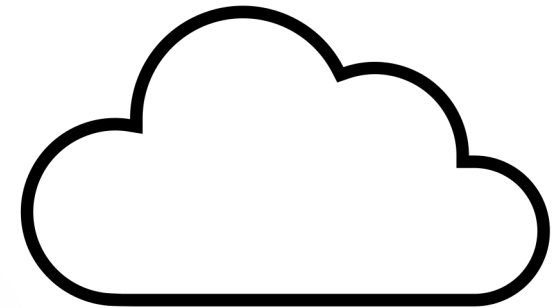
Access

Direct access service for partners and premium customers, with QuEra support

On-premises **managed service** through G-QuAT/ABCI-Q.



Public **cloud service**.



What would you like to do with Gemini?

